

Technical Documentation: Portfolio SPA

AWS Serverless Deployment Architecture

1. High-Level Architecture

This architecture leverages AWS serverless services to host a Single Page Application (SPA) with global reach, high security, and zero server management.

```
graph LR
    User([User]) -- HTTPS --> CF[CloudFront CDN]
    subgraph "AWS Cloud"
        CF -- "OAC Signed Request" --> S3[(S3 Private Bucket)]
        ACM[AWS Certificate Manager] --- CF
    end
    subgraph "CI/CD Pipeline"
        GH[GitHub Repo] -- Webhook --> CP[CodePipeline]
        CP --> CB[CodeBuild]
        CB -- "Sync Assets" --> S3
        CB -- "Invalidate Cache" --> CF
    end
```

2. Deployment Workflow (CI/CD)

The workflow automates the transition from code commit to production delivery.

```
sequenceDiagram
    participant Dev as Developer
    participant GH as GitHub
    participant CP as CodePipeline
    participant CB as CodeBuild
    participant S3 as S3 Bucket
    participant CF as CloudFront
```

```
Dev->>GH: git push origin main
GH->>CP: Trigger Pipeline
CP->>CB: Start Build Environment
Note over CB: npm install && npm run build
CB->>S3: aws s3 sync ./dist
CB->>CF: Create Invalidation (/*)
CP-->>Dev: Deployment Successful
```

3. Scaling Strategy

- **Global Delivery:** **CloudFront** uses 400+ Edge Locations to cache content closer to users, scaling automatically to handle millions of requests.
- **Storage Resilience:** **Amazon S3** provides 99.999999999% durability and handles request scaling without manual intervention.
- **On-Demand Builds:** **CodeBuild** scales concurrently, spinning up fresh containers for every build.

4. Security Controls

Control	Implementation	Threat Mitigated
Origin Access Control (OAC)	S3 Policy permits only CloudFront access.	Direct S3 URL access and data scraping.
Encryption	ACM Managed SSL/TLS Certificates.	Man-in-the-Middle (MITM) attacks.
Least Privilege	IAM Roles for CodeBuild and Pipeline.	Unauthorized AWS resource access.
Private Origin	S3 Block Public Access (BPA) enabled.	Accidental data exposure via public S3.